

Experiment 17

Understand Linear Regression for a given data set.

Program:

```
import numpy as np
import pandas as pd

df=pd.read_csv('Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()

features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(
    features, label, test_size=0.2, random_state=23
)

from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)

print(model.score(x_train,y_train))
print(model.score(x_test,y_test))
print(model.coef_)
print(model.intercept_)

import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: "
      .format(yr_of_exp,Salary))
```

Output

```
Output:
Training score: 0.9603182547438908
Testing score: 0.9184170849214232
Coefficient: [[9281.30847068]]
Intercept: [27166.73682891]

Enter Years of Experience: 44
Estimated Salary for 44.0 years of experience is
[[435544.30953887]]
```